Course Code: BCE-C621
Course Name: DISTRIBUTED SYSTEMS

MM: 100 **Time: 3 Hr.** **L T P** **3 1 0**	**Sessional:30** **ESE:70** **Credit :4**

Prerequisites:	Familiarity with the design and analysis of sequential algorithms, knowledge of basic computer organization and elementary operating systems concepts are required.
Objectives:	• To understand the foundations of distributed systems. • To learn issues related to clock Synchronization and the need for global state in distributed systems. • To learn distributed mutual exclusion and deadlock detection algorithms. • To understand the significance of agreement, fault tolerance and recovery protocols in Distributed Systems. • To learn the characteristics of peer-to-peer and distributed shared memory systems.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.

Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	Characterization of Distributed Systems-Introduction-Examples-Resource Sharing and the Web-Challenges. System Models-Architectural-Fundamental. Inter process Communication- Introduction-API for Internet Protocols-External data representation and marshalling--Client-server Communication-Group communication- Case study: Inter process Communication in UNIX.Transparency in Distributed Systems (Access, Location, Migration, etc.)	**08**	PO1/ PO2	PSO1/ PSO2
Module-2	Distributed Objects and Remote Invocation-Introduction-Communication between distributed objects-Remote procedure calls-Events and notifications-Case study: Java RMI. Operating System Support-Introduction-OS layer-Protection-Processes and threads- Communication and invocation OS architecture.Sockets and Streams,Quorum-based Protocol	**08**	PO2/ PO3	PSO1/ PSO2
Module-3	Distributed File Systems-Introduction-File service architecture-Case Study: Sun Network File System-Enhancements and further developments. Name Services-Introduction-Name Services and the Domain Name System-Directory. Services-Case Study: Global Name Service.	**08**	PO1/ PO2/ PO3	PSO1/ PSO2
Module-4	Time and Global States-Introduction-Clocks, events and process states-Synchronizing physical clocks-Logical time and logical clocks-Global States-Distributed debugging. Coordination and Agreement-Introduction-Distributed mutual exclusion-Elections-Multicast Communication-Consensus and related problems. Authentication in distributed systems: Protocols based on symmetric cryptosystems, Protocols based on asymmetric cryptosystems, Password-based authentication, and Authentication protocol failures.	**08**	PO1/ PO3/ PO12	PSO1/ PSO2

Module-5	Distributed Shared Memory-Introduction-Design and implementation issues-Sequential consistency and Ivy case study Release consistency and Munin case study-Other consistency models. CORBA Case Study-Introduction-CORBA RMI-CORBA services,Cloud-Native Distributed Systems (Containers, Kubernetes), Microservices Architecture in Distributed Systems, Middleware and Message Queues (Kafka, RabbitMQ)	**08**	PO1/ PO2/ PO3	PSO1/ PSO2
Total No. of Hours		**40**		

Learning Outcomes:	• Understanding of Distributed architecture, CORBA, RMI. • Understanding reliable services to geographically dispersed users. The focus is networks, server architecture, protocols, security, resiliency, and scalability. • Illustrate distributed file system, time and global state. • Understand properties of key algorithms used in concurrent systems such as mutual exclusion. Its focus is the application of formal techniques.

Suggested books:

S. No.	Name of Authors /Books /Publisher
1.	George Coulouris, Jean Dollimore, Tim Kindberg, "Distributed Systems: Concepts and Design", 4th Edition, Pearson Education, 2005.
2.	A.tS. Tanenbaum and M. V. Steen, "Distributed Systems: Principles and Paradigms", Second Edition, Prentice Hall, 2006.
3.	M.L.Liu, “Distributed Computing Principles and Applications”, Pearson Addison Wesley, 2004.
4.	MukeshSinghal, “Advanced Concepts In Operating Systems”, McGrawHill Series in Computer Science, 1994.
5.	Nancy A. Lynch, "Distributed Algorithms", The Morgan Kaufmann Series in Data Management System, Morgan Kaufmann Publishers, 2000.

CO-PO/PSO MAPPING

Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	√	√											√	√
CO2	√	√	√										√	√
CO3	√	√	√									√	√	√
CO4	√	√										√	√	√